

# Install mongosh on Windows

*MongoDB Shell setup for the NoSQL / MongoDB session — student handout*

mongosh is the MongoDB Shell — the tool you type database commands into. You need it to run queries against our cloud database (MongoDB Atlas). Two ways to install it on Windows are covered below: the MSI installer (simplest, recommended) and the ZIP archive (manual, useful if you can't run installers on your machine). Pick ONE method, not both.

**What you are installing:** only mongosh — the shell/client. You do NOT need to install a MongoDB server, because our database lives in the cloud on Atlas. This keeps the install small and fast.

## Method 1 — MSI Installer (simplest, recommended)

1. In a browser, go to: [mongodb.com/try/download/shell](https://mongodb.com/try/download/shell)
2. Under 'Platform', choose Windows 64-bit (8.1+) — the package format should say msi.
3. Click Download — it saves a file like mongosh-2.x.x-win32-x64.msi into your Downloads folder.
4. Double-click the downloaded .msi file.
5. Follow the installation wizard — choose the 'Complete' install type, then Finish. The installer adds mongosh to your PATH automatically.

**Important:** close any Command Prompt or PowerShell window you already had open, and start a NEW one. Windows only picks up the updated PATH in newly opened terminal windows — an old window still won't recognize mongosh.

## Method 2 — ZIP Archive (manual, no installer)

Use this if you can't run .msi installers on your machine (e.g. restricted work/college laptop).

1. In a browser, go to: [mongodb.com/try/download/shell](https://mongodb.com/try/download/shell)
2. Under 'Platform', choose Windows 64-bit (8.1+) — the package format should say zip.
3. Click Download — it saves a file like mongosh-2.x.x-win32-x64.zip into your Downloads folder.
4. Right-click the zip file -> 'Extract All...' -> extract it somewhere permanent, e.g. C:\mongosh (not Downloads, so it doesn't get deleted later).
5. Inside the extracted folder, find the bin subfolder (e.g. C:\mongosh\mongosh-2.x.x-win32-x64\bin) — this contains mongosh.exe. Copy this full folder path.

### Now add that bin folder to your PATH:

1. Click Start, type 'environment variables', and open 'Edit the system environment variables'.
2. Click the 'Environment Variables...' button.
3. Under 'User variables' (top half), select the row named Path, then click Edit.
4. Click New, and paste the bin folder path you copied (e.g. C:\mongosh\mongosh-2.x.x-win32-x64\bin).
5. Click OK on all three open windows to save.

**Important:** close any Command Prompt or PowerShell window you already had open, and start a NEW one — the PATH change only applies to windows opened after you saved it.

## Verify the install

In a NEW Command Prompt or PowerShell window, run:

```
mongosh --version  
# e.g. 2.x.x
```

If you see a version number, mongosh is installed correctly. If you instead see 'mongosh is not recognized as an internal or external command', see Troubleshooting below.

## Connect to the class Atlas cluster

Once you have your own Atlas cluster and database user set up (we do this together at the start of the session), get your connection command from Atlas:

1. In Atlas, click 'Connect' on your cluster.
2. Choose 'Shell'.
3. Click 'I have the MongoDB Shell installed'.
4. Copy the command shown — it looks like the one below.

```
mongosh "mongodb+srv://yourcluster.xxxxx.mongodb.net/" --apiVersion 1  
--username youruser  
  
# it will prompt: Enter password: ****  
# type (or paste) your DATABASE USER password, then press Enter  
# (the password is hidden as you type – that is normal)
```

When it connects, your prompt changes to something like this. The word 'Atlas' means you are connected to the cloud cluster:

```
Atlas atlas-abc123-shard-0 [primary] test>
```

Test it with:

```
db.version()  
# shows the database version – you're ready!
```

**Most common mistake:** the mongosh "mongodb+srv://..." command is a TERMINAL command. Run it at your normal Command Prompt / PowerShell prompt, NOT inside mongosh. If your prompt already shows 'test>' or 'Atlas...>', you are inside mongosh — the connect command will fail there with a 'SyntaxError'. Type exit first, then run it from a normal terminal window.

## Finding your user & resetting the password (if you forgot it)

Atlas never shows an existing password again after it's created — it's stored securely. If you don't remember it, you don't recover it, you RESET it. That's normal and takes 30 seconds.

### See which database users exist

1. In Atlas, open the left sidebar -> 'Database & Network Access' (or 'Security' -> 'Database Access').
2. Click the 'Database Users' tab.
3. Every user for your project is listed here with its username. This is the username you use in the connect command (`--username ...`).

### Reset the password

1. On your user's row, click 'Edit'.
2. Click 'Edit Password'.
3. Click 'Autogenerate Secure Password' (or type your own) and COPY it somewhere safe.
4. Click 'Update User' and wait a few seconds for it to apply.
5. Use this new password the next time mongosh asks 'Enter password:'.

**Tip: use a simple password (letters + numbers only).** Special characters like @ : / # must be URL-encoded in a connection string and are a common cause of 'authentication failed'. A plain letters-and-numbers password avoids all of that.

## Troubleshooting — the common errors

### 'mongosh' is not recognized as an internal or external command

- You're in an OLD terminal window — close it and open a brand new Command Prompt or PowerShell window.
- MSI method: reinstall and confirm you left 'Complete' selected — this ensures PATH is set automatically.
- ZIP method: double-check the exact bin folder path was added under 'User variables' -> Path (not a typo, not missing the \bin at the end).
- As a last resort, run mongosh using its full path once to confirm it works at all, e.g.  
`C:\mongosh\mongosh-2.x.x-win32-x64\bin\mongosh.exe --version`

### 'Authentication failed' / 'bad auth'

- Wrong database-user password. In Atlas: Database Access -> Edit your user -> reset the password, then reconnect.
- If your password has special characters (@ : / etc.), they must be URL-encoded in the connection string. Easiest fix: set a password with only letters and numbers.

### 'connection timed out' / 'SSL alert number 80' / IP not allowed

- This means your IP isn't on the cluster's allow-list — the single most common connection error. It is NOT a password problem.
- Fix: Atlas -> Database & Network Access -> IP Access List -> '+ Add IP Address'.
- In the box, type 0.0.0.0/0 (allow from anywhere) and click Confirm — simplest for class use.
- Wait ~1 minute for the entry's status to turn green (Active), then re-run the mongosh command.
- Note: this is set on YOUR OWN cluster — doing it on someone else's cluster won't help you.

### 'querySrv ENOTFOUND'

- The connection string was mistyped or partly copied. Re-copy the whole `mongodb+srv://...` command from Atlas -> Connect -> Shell.

**Note:** mongosh only connects to a database — it doesn't create one on your computer. Our database lives on Atlas (cloud), so that's all you need. You do not need to install a local MongoDB server for this course.